

Vijay Mohanram Iyer

📍 Tennesseeallee 20, Karlsruhe, Baden-Württemberg, Germany

✉ iyervijay99@gmail.com

☎ +49-17667345305

🌐 [Vijay's Portfolio](#)

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SUMMARY

Graduate engineer specializing in intelligent safety systems and health applications powered by computer vision and machine learning. Skilled in implementing and optimizing deep learning architectures, transformer-based models, and multimodal pipelines, with expertise in model training & Fine-Tuning, transfer learning, signal processing, and real-time deployment for user-centered AI solutions..

PROFESSIONAL EXPERIENCE

1. Forschungszentrum Informatik *September 2025 – Present* **Research Assistant**

- Optimize and explore advanced generative models through transfer learning, ensemble methods, latent-diffusion techniques, and 3D geometric facemesh reconstruction for forged video rendering and region-specific analysis.
- Conduct literature review and work towards research publication.

2. TecoLab *March 2023 – Present* **Working Student**

- Sensor optimization for open-earables with TinyML and quantized neural networks for on-device inference.
- ML4Print: Designed a CNN based pipeline to classify printing techniques and paper substrates, using saliency maps for explainability and reducing manual forensic effort by 80%.
- Applied regression models and data integration to simulate liquid thermal behavior in industrial valves, improving predictive maintenance and system reliability.

3. Forschungszentrum Informatik *February 2025 – August 2025* **Master Thesis (Grade: 1,0)**

- Developed an end-to-end deep learning pipeline using rPPG signals for artifact evaluation in vital signals and Benchmarked under real-world conditions.
- Designed a Fusion Model combining Vision Transformer and CNN features via a custom Fusion-Head for DeepFake classification.

4. Access@KIT *March 2023 – May 2023* **Working Student**

- Implemented a diffusion-based text-to-speech framework with phoneme alignment and acoustic features, generating high-quality, natural speech to support visually impaired users.

5. Accenture India Pvt. Ltd. *February 2022 – April 2022* **Associate Software Engineer**

- Maintained IBM Mainframe servers with batch job scheduling and performance monitoring COBOL scripts.

6. Accur Digitus
Web Developer Intern

June 2021 – January 2022

- Developed responsive web applications using React.JS and Tailwind CSS, integrated scalable RESTful APIs, and enhanced user experience, resulting in over 30% increase in engagement.

EDUCATION

M.Sc. in Electrical Engineering and Information Technology Karlsruhe Institute of Technology GPA: 2.3	<i>May 2022 – July 2025</i>
Bachelor of Engineering in Electronics Engineering University of Mumbai, India GPA: 2.8	<i>August 2017 – May 2021</i>

PROJECTS

- 1. Non-Contact based vital Extraction for Health Monitoring with Attention Mechanisms (FZI)** *February 2025 – August 2025*
Utilized POS and CWT to extract pulsatile components from facial videos, improving robustness for non-contact vital sign monitoring in telemedicine.
Technology: Python, PyTorch, OpenCV, NumPy, Matplotlib, ViT, rPPG, GenAI
- 2. CamCussion (Zeiss Innovation Hub)** *November 2023 – July 2024*
Utilized pupil segmentation, gaze tracking, and saccadic velocity estimation to analyze pupil dilation and eye movements in real time contributing for accurate concussion diagnosis
Technology: Python, OpenCV, Dlib, PyQt, image processing, PyTorch, NumPy, ML
- 3. Self-Driving Car using LIDAR** *September 2022 – February 2023*
Developed real-time 3D obstacle detection by voxelizing LiDAR point clouds into ViT tokens and applying Multi-head attention for improved accuracy.
Technology: Python, PyTorch, Open3D, NumPy, ViT, LiDAR processing
- 4. Real-Time Car Accident Alert System** *January 2022 – June 2022*
Built ESP32-based embedded LSTM system with TensorFlow Lite for real-time crash detection and automated SMS/email alerts with GPS location.
Technology: Python, TensorFlow Lite, ESP32, LSTM, Embedded C

TECHNICAL SKILLS

- **Programming Languages:** Python, C++, JavaScript, SQL, Node.js, HTML, CSS
- **Libraries/Frameworks:** OpenCV, Qt, TensorFlow, PyTorch, pandas, NumPy, Matplotlib, scikit-learn, React.js
- **ADAS:** Sensor fusion, camera and LiDAR calibration, Point cloud, object detection & tracking
- **Embedded Systems:** ARM Cortex-M, ESP32, STM32, Arduino, Raspberry Pi, RTOS, Embedded C/C++
- **Tools & Cloud Platforms:** Git, Docker, MATLAB, Linux, Carla, conda, LaTeX, Jupyter Notebook, Azure, AWS
- **Spoken Languages:** English C1, German A2 (Learning)

CORE COMPETENCIES

- **Technical Leadership:** Project Development, Cross-functional Collaboration, Innovative mindset.
- **Problem Solving:** Critical Thinking, Performance Optimization, Debugging.
- **Communication:** Active Listening, Documenting, Knowledge Transfer.